Statistics

Chapter 14 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 2

PROBLEM STATEMENT

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Solution A

Step 1

(a) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$df_{total} = 29$$

The total degrees of freedom is the sample size decreased by 1, thus the sample size is obtained by increasing the total degrees of freedom increased by 1.

$$n = df_{total} + 1 = 29 + 1 = 65$$

Step 2

(b) The estimate of the constant a is given in the row "Constant" and in the column "Coefficients" of the computer output:

$$a = 90.6190$$

The estimate of the coefficient b_1 of the variable x is given in the row "Slope" and in the column "Coefficients" of the computer output:

$$b = -0.9401$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + bx = 90.6190 - 0.9401x$$

The estimated value of the dependent variable is 90.6190 when the independent value is 0.

The dependent value decreases on average by 0.9401 per unit of the dependent variable.

Step 3

(c) Evaluate the multiple regression equation from part (b) at $x = 10$:

$$\hat{y} = a + bx = 90.6190 - 0.9401(10) = 81.2180$$

Thus the value of the dependent variable is estimated to be 81.2180.

Step 4

(d) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 129.7275$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 380.1667$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{129.7275}{380.1667} \approx 0.3412 = 34.12\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

34.12% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 5

(e) The estimate of the coefficient b_1 of the variable x is given in the row "Slope" and in the column "Coefficients" of the computer output:

$$b = -0.9401$$

Since the slope $b = -0.9401$ is negative, there is a negative relationship between the variables and thus the correlation coefficient needs to be negative as well.

The correlation coefficient is then the negative square root of the coefficient of determination.

$$r = -\sqrt{r^2} = -\sqrt{0.3412} \approx$$

Step 6

Given claim: Negative association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \geq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho < 0 \text{ (There is a negative linear correlation)}$$

Step 7

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.10$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 30 - 2 = 28$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.701$$

The rejection region then contains all values smaller than -1.701 and thus we reject H_0 when the t-value is smaller than -1.701.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.5841\sqrt{28-2}}{\sqrt{1-(-0.5841)^2}} \approx -3.669$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = --3.669 < -1.701 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a negative association between the variables.

🎯 Result

(a) $n = 30$

(b) $\hat{y} = -0.6190 - 0.9401x$

(c) 81.2180

(d) 34.12% of the variation in the dependent variable is explained by the variation in the independent variables.

(e) $r = -0.5841$, There is sufficient evidence to support the claim of a negative association between the variables.

PROBLEM STATEMENT

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Solution A

Step 1

(a) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$df_{total} = 29$$

The total degrees of freedom is the sample size decreased by 1, thus the sample size is obtained by increasing the total degrees of freedom increased by 1.

$$n = df_{total} + 1 = 29 + 1 = 30$$

Step 2

(b) The regression degrees of freedom is given in the row "Regression" and in the column "DF" of the ANOVA table:

$$df_R = 4$$

The regression degrees of freedom represent the number of independent variables in the study and thus the study contains 4 independent variables in this case.

Step 3

(c) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 227.0928$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 380.1667$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{227.0928}{380.1667} \approx 0.5974 = 59.74\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

59.74% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 4

(d)

$$\alpha = \text{Significance level} = 0.01$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "p-value" of the ANOVA table:

$$P = 0.000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.000 < 0.01 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 5

(e)

$$\alpha = \text{Significance level} = 0.05$$

The P-values are given in the column "p-value" of the regression output.

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$\begin{aligned} x_1 \quad & P = 0.010 < 0.05 \Rightarrow \text{ Reject } H_0 \\ x_2 \quad & P = 0.690 > 0.05 \Rightarrow \text{ Fail to reject } H_0 \\ x_3 \quad & P = 0.006 < 0.05 \Rightarrow \text{ Reject } H_0 \\ x_4 \quad & P = 0.030 < 0.05 \Rightarrow \text{ Reject } H_0 \end{aligned}$$

There is not sufficient evidence to support the claim that the coefficient of X_2 is nonzero, while there is sufficient evidence to support the claim that all other coefficients are nonzero.

Since the coefficient of X_2 appears to be approximately zero, we would consider eliminating variable X_2.

Step 6

(f) We start by deleting the independent variable with the highest P-value, which is the independent variable X_1.

We determine the regression output for the model with independent variable X_1 deleted. If any of the independent variables still has a P-value greater than 0.05 in the new regression model, then we proceed with deleting this variable from the model as well.

Result

(a) $n = 30$

(b) 4 independent variables

(c) 59.74%

(d) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(e) Yes, X_2